

PROJECT REPORT

Project Title: FitBuddy - AI Workout Generator **Program:** Naan Mudhalvan (SmartBridge)

1. Team Details

- **Team ID:** SWTID-2026-2770
- **Team Size:** 4
- **Team Leader:** Manikandaprabu
- **Team Members:** Muthuvelan R, Mohamed Kanifa P, Syed Sulaiman

2. Abstract

FitBuddy is a personalized AI-driven fitness planner designed to dynamically generate customized 7-day workout plans and nutrition tips. Recognizing that standard fitness plans do not work for everyone, this application leverages Generative AI to tailor workout routines based on an individual's age, weight, specific fitness goals (e.g., weight loss), and preferred intensity levels. Built on a fast and modern web framework, FitBuddy provides a seamless user experience from data input to real-time AI generation and database storage.

3. Project Objectives

- To develop a responsive web application that collects user fitness profiles and goals.
- To integrate Google Gemini Generative AI to create customized, day-by-day workout plans instantly.
- To store user data and generated plans persistently using a relational database for future retrieval and admin monitoring.
- To implement seamless fallback mechanisms (timeout handling) ensuring a continuous user experience even during API latency.

4. Technology Stack

- **Programming Language:** Python 3
- **Backend Framework:** FastAPI
- **Database & ORM:** SQLite, SQLAlchemy
- **AI Integration:** Google Gemini API (gemini-3.5-flash-lite)
- **Frontend:** Jinja2 Templates, HTML, CSS
- **Server:** Uvicorn

5. System Architecture & Flowchart

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The application follows a linear client-server architecture with external API integration:

1. **User Input:** The user submits their details (Name, Age, Weight, Goal, Intensity) via the Jinja2 HTML form.
2. **FastAPI Route:** The backend receives the POST request and temporarily stores the data.
3. **Gemini API Processing:** A crafted prompt containing the user's data is sent to the Google Gemini API.
4. **Response Generation:** The AI returns a structured 7-day plan and a separate nutrition tip.
5. **Database Storage:** SQLAlchemy commits the user profile and the original AI-generated plan into the SQLite database.
6. **Jinja2 UI:** The final customized plan is rendered elegantly back to the user on the result page.

Data Flow: User UI → FastAPI (routes.py) → Gemini API → SQLite (database.py) → Result UI

6. Key Code Snippets

A. Database Schema (database.py)

Python:

```
class User (Base):
```

```
    _tablename = "users"
```

```
    " id = Column(Integer, primary_key=True, index=True)
```

```
    name = Column(String)
```

```
    age = Column (Integer)
```

```
    weight = Column (Float) goal = Column(String)
```

```
    intensity = Column (String)
```

```
class WorkoutPlan (Base):
```

```
    tablename = "workout_plans"
```

```
    id = Column (Integer, primary_key=True, index=True)
```

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```
user_id = Column(Integer)

original_plan = Column(String)

**B. Gemini AI Integration (gemini_generator.py)

import google.generativeai as genai

import os

genai.configure(api_key=os.environ.get("GOOGLE_API_KEY"))

model = genai.GenerativeModel("gemini-3.5-flash-lite")

def generate_workout_gemini(user_input):

    prompt = f"Create a structured 7-day workout plan for {user_input['goal']} at {user_input['intensity']} intensity."

    try:

        response = model.generate_content(prompt, request_options={"timeout": 60})
        return response.text

    except Exception as e:

        return f"Error: {e}"

**C. FastAPI Routing (routes.py)

@router.post("/generate-workout", response_class=HTMLResponse)

def generate_workout(request: Request, user_id: int = Form(...), goal: str = Form(...), intensity: str = Form(...)):

    # Database Logic

    db = SessionLocal()

    # AI Generation

    user_input = {"goal": goal, "intensity": intensity}

    plan = generate_workout_gemini(user_input)

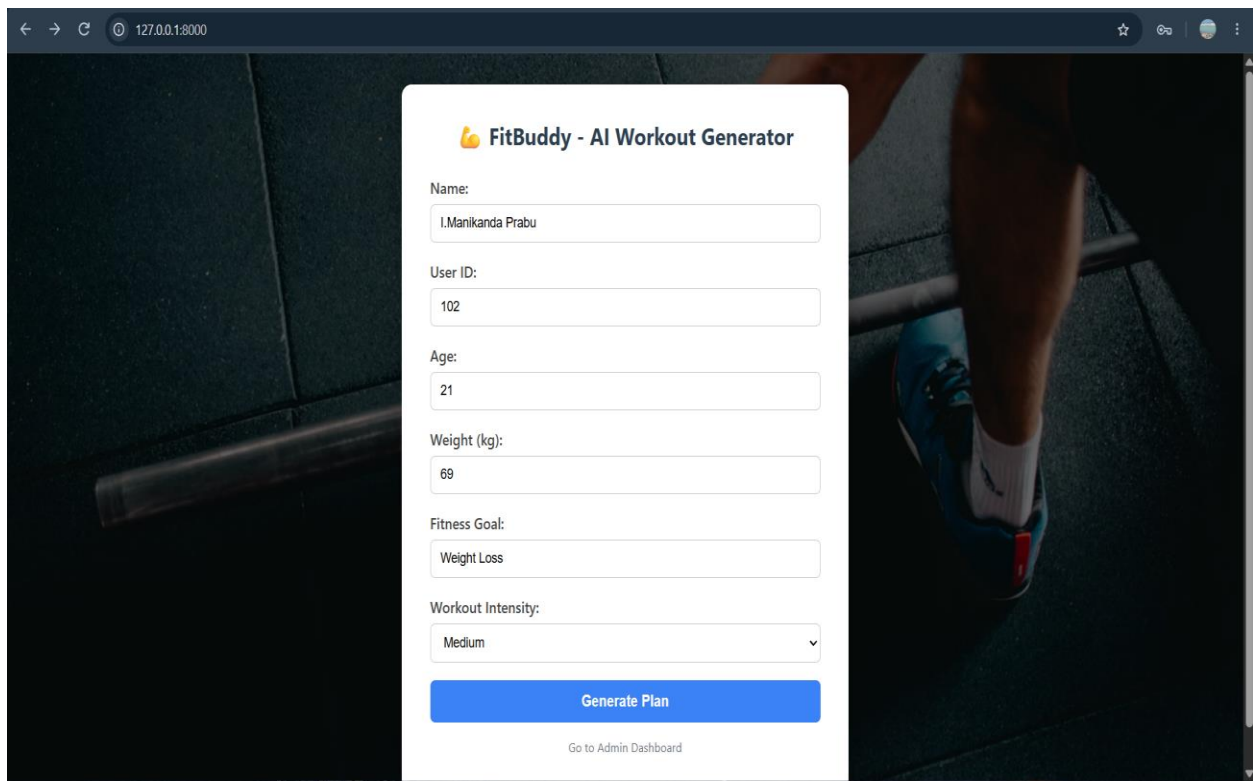
    # Save & Render

    return templates.TemplateResponse(request=request, name="result.html", context={"workout_plan": plan})
```

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7. Output Screenshots:

- Screenshot 1: Home Page / User Input Form



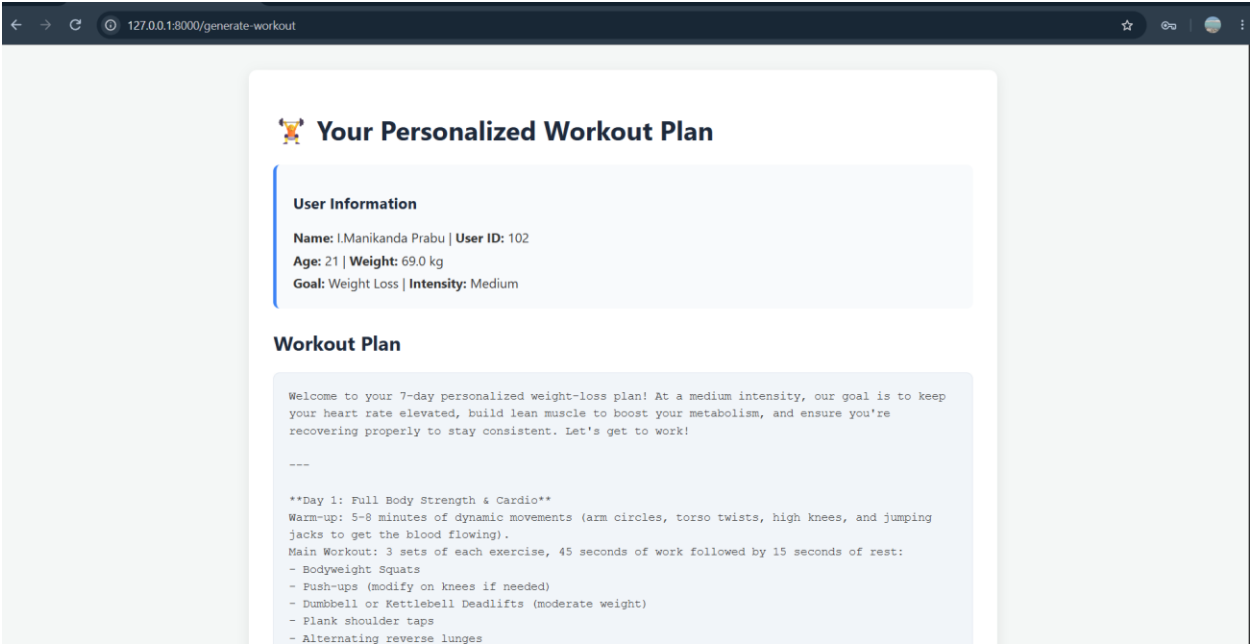
The screenshot shows a web browser window displaying the 'FitBuddy - AI Workout Generator' user input form. The form is centered on a dark background featuring a person's legs and feet in blue sneakers. The form fields are as follows:

- Name:** Input field containing 'I.Manikanda Prabu'.
- User ID:** Input field containing '102'.
- Age:** Input field containing '21'.
- Weight (kg):** Input field containing '69'.
- Fitness Goal:** Input field containing 'Weight Loss'.
- Workout Intensity:** Dropdown menu with 'Medium' selected.

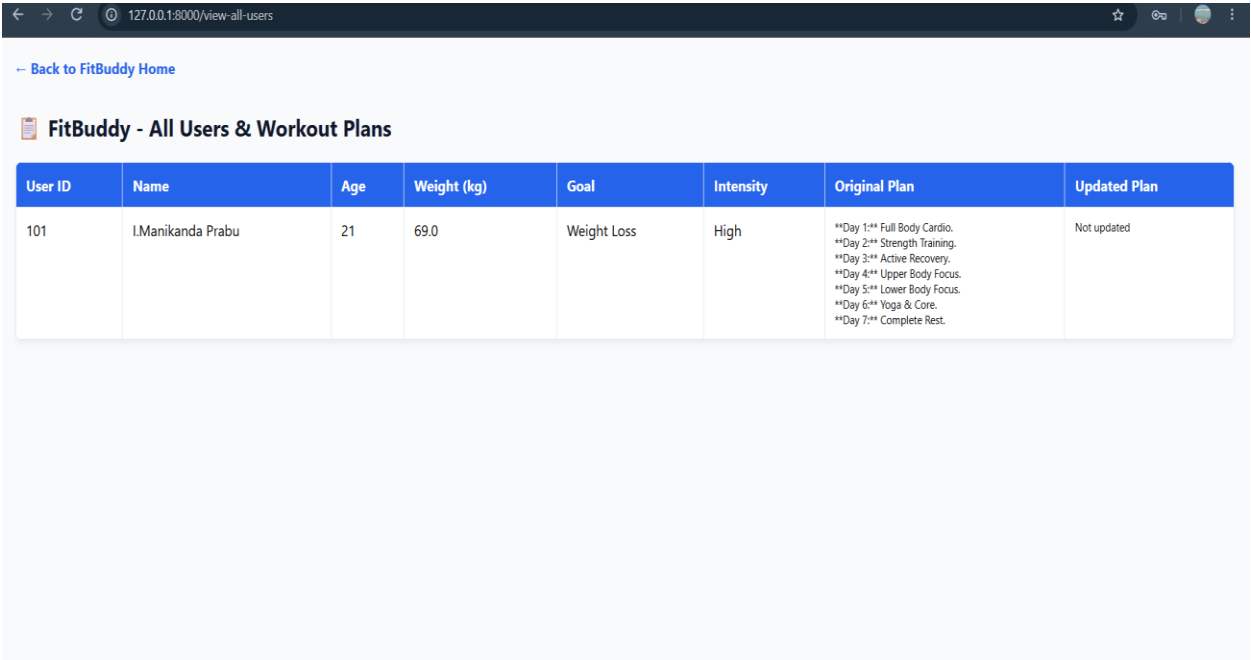
Below the form fields is a blue button labeled 'Generate Plan'. At the bottom of the form, there is a link that says 'Go to Admin Dashboard'.

- Screenshot 2: Generated 7-Day Workout Plan

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• Screenshot 3: Admin Dashboard (All Users View)



8. Conclusion:

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The FitBuddy project successfully demonstrates the integration of modern web frameworks (FastAPI) with advanced Generative AI (Google Gemini). By automating personalized fitness planning, the application serves as a scalable prototype for health-tech solutions. The project effectively met all Naan Mudhalvan requirements, showcasing robust backend routing, database management, and error handling.